## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1 Product identifier

| | |
|---|---|
| **Trade name** | GLUFOSINATE-AMMONIUM SL 150I G |
| **Product code (UVP)** | 85327623 |

### 1.2 Relevant identified uses of the substance or mixture and uses advised against

| | |
|---|---|
| **Use** | Herbicide |

### 1.3 Details of the supplier of the safety data sheet

| | |
|---|---|
| **Supplier** | Bayer AG<br>Kaiser-Wilhelm-Allee 1<br>51373 Leverkusen<br>Germany |
| **Telefax** | +49(0)2173-38-7394 |
| **Responsible Department** | Substance Classification & Registration<br>+49(0)2173-38-3409 (during business hours only)<br>Email: BCS-SDS@bayer.com |

### 1.4 Emergency telephone no.

| | |
|---|---|
| **Emergency telephone no.** | Global Incident Response Hotline (24h)<br>+1 (760) 476-3964 (Company 3E for Bayer AG, Crop Science Division) |

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1 Classification of the substance or mixture

**Classification in accordance with Regulation (EC) No 1272/2008 on classification, labelling and packaging of substances and mixtures, as amended.**

Acute toxicity: Category 3
H311 Toxic in contact with skin.

Acute toxicity: Category 4
H302 Harmful if swallowed.

Reproductive toxicity: Category 1B
H360Fd May damage fertility. Suspected of damaging the unborn child.

Serious eye damage: Category 1
H318 Causes serious eye damage.

Specific target organ toxicity - repeated exposure: Category 2
H373 May cause damage to organs (Nervous system) through prolonged or repeated exposure if swallowed.

Chronic aquatic toxicity: Category 3
H412 Harmful to aquatic life with long lasting effects.

### 2.2 Label elements

**Labelling in accordance with Regulation (EC) No 1272/2008 on classification, labelling and packaging of substances and mixtures, as amended.**

Hazard label for supply/use required.

**Hazardous components which must be listed on the label:**

- Glufosinate ammonium

**Signal word**: Danger

**Hazard statements**

| | |
|---|---|
| H302 | Harmful if swallowed. |
| H311 | Toxic in contact with skin. |
| H318 | Causes serious eye damage. |
| H360Fd | May damage fertility. Suspected of damaging the unborn child. |
| H373 | May cause damage to organs (Nervous system) through prolonged or repeated exposure if swallowed. |
| H412 | Harmful to aquatic life with long lasting effects. |
| EUH401 | To avoid risks to human health and the environment, comply with the instructions for use.<br>Restricted to professional users. |

**Precautionary statements**

| | |
|---|---|
| P201 | Obtain special instructions before use. |
| P280 | Wear protective gloves/ protective clothing/ eye protection/ face protection. |
| P305 + P351 + P338 | IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. |
| P310 | Immediately call a POISON CENTER/doctor/ physician. |
| P501 | Dispose of contents/container in accordance with local regulation. |

**2.3 Other hazards**

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

**3.2 Mixtures**

**Chemical nature**

Soluble concentrate (SL)
Glufosinate-ammonium 150 g/l

**Hazardous components**

Hazard statements according to Regulation (EC) No. 1272/2008

| Name | CAS-No. / EC-No. / REACH Reg. No. | Classification REGULATION (EC) No 1272/2008 | Conc. [%] |
|---|---|---|---|
| Glufosinate ammonium | 77182-82-2<br>278-636-5 | Acute Tox. 4, H302<br>Repr. 1B, H360Fd<br>Acute Tox. 4, H332<br>Acute Tox. 4, H312<br>STOT RE 2, H373 | 13,51 |
| Alkylethersulfate, sodium | 68891-38-3 | Eye Dam. 1, H318 | > 35 – < 50 |

| | | | |
|---|---|---|---|
| salt | 500-234-8 | Skin Irrit. 2, H315<br>Aquatic Chronic 3, H412 | |
| 1-Methoxy-2-propanol | 107-98-2<br>203-539-1<br>01-2119457435-35-xxxx | Flam. Liq. 3, H226<br>STOT SE 3, H336 | > 5 – < 10 |

**Further information**

Substances for which there are Community workplace exposure limits:
1-Methoxy-2-propanol (107-98-2)

For the full text of the H-Statements mentioned in this Section, see Section 16.

## SECTION 4: FIRST AID MEASURES

### 4.1 Description of first aid measures

**General advice**
Move out of dangerous area. Place and transport victim in stable position (lying sideways). Remove contaminated clothing immediately and dispose of safely. Keep under medical supervision for at least 48 hours.

**Inhalation**
Move to fresh air. Keep patient warm and at rest. Call a physician or poison control center immediately.

**Skin contact**
Wash off immediately with soap and plenty of water. Call a physician or poison control center immediately.

**Eye contact**
Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present, after the first 5 minutes, then continue rinsing eye. Call a physician or poison control center immediately.

**Ingestion**
Do NOT induce vomiting. Rinse mouth. Call a physician or poison control center immediately.

### 4.2 Most important symptoms and effects, both acute and delayed

**Symptoms**
Nausea, Vomiting, Diarrhoea, Abdominal pain, Hypotension, Tremors, Muscular weakness, Tachycardia, Convulsions, Unconsciousness, Coma, Respiratory failure

Symptoms may be delayed.

Symptoms and hazards refer to effects observed after intake of significant amounts of the active ingredient(s).

### 4.3 Indication of any immediate medical attention and special treatment needed

**Risks**
Watch victim for at least 48 hours because of possible delayed signs of poisoning.

**Treatment** Appropriate supportive and symptomatic treatment as indicated by the patient's condition is recommended. In case of ingestion gastric lavage should be considered in cases of significant ingestions only within the first 2 hours. However, the application of activated charcoal and sodium sulphate is always advisable. Forced alkaline diuresis and hemodialysis may be considered. There is no specific antidote. In case of convulsions, a benzodiazepine (e.g. diazepam) should be given according to standard regimens. If not effective, phenobarbital may be used. Contraindication: atropine. Oxygen or artificial respiration if needed. Keep respiratory tract clear. ECG - monitoring (Electrocardiogram). EEG - monitoring (Electroencephalogram). Monitor: respiratory, cardiac, kidney, liver and central nervous system. Keep under medical supervision for at least 48 hours.

## SECTION 5: FIREFIGHTING MEASURES

### 5.1 Extinguishing media

**Suitable** Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.

**Unsuitable** High volume water jet

**5.2 Special hazards arising from the substance or mixture** In the event of fire the following may be released:, Hydrogen cyanide (hydrocyanic acid), Carbon monoxide (CO), Carbon dioxide (CO2), Oxides of phosphorus, Nitrogen oxides (NOx)

### 5.3 Advice for firefighters

**Special protective equipment for firefighters** In the event of fire and/or explosion do not breathe fumes. In the event of fire, wear self-contained breathing apparatus.

**Further information** Remove product from areas of fire, or otherwise cool containers with water in order to avoid pressure being built up due to heat. Contain the spread of the fire-fighting media. Do not allow run-off from fire fighting to enter drains or water courses.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1 Personal precautions, protective equipment and emergency procedures

**Precautions** Avoid contact with spilled product or contaminated surfaces. Use personal protective equipment.

**6.2 Environmental precautions** Do not allow to get into surface water, drains and ground water.

### 6.3 Methods and materials for containment and cleaning up

**Methods for cleaning up** Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Clean contaminated floors and objects thoroughly, observing environmental regulations.

**6.4 Reference to other sections**

Information regarding safe handling, see section 7.
Information regarding personal protective equipment, see section 8.
Information regarding waste disposal, see section 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1 Precautions for safe handling

**Advice on safe handling**

Use only in area provided with appropriate exhaust ventilation.

**Advice on protection against fire and explosion**

Keep away from heat and sources of ignition.

**Hygiene measures**

Avoid contact with skin, eyes and clothing. Keep working clothes separately. Wash hands immediately after work, if necessary take a shower. Remove soiled clothing immediately and clean thoroughly before using again. Garments that cannot be cleaned must be destroyed (burnt).

### 7.2 Conditions for safe storage, including any incompatibilities

**Requirements for storage areas and containers**

Store in original container. Keep containers tightly closed in a dry, cool and well-ventilated place. Store in a place accessible by authorized persons only. Keep away from direct sunlight. Protect from freezing.

**Suitable materials**

Coex HDPE/EVOH/HDPE
HDPE (high density polyethylene)

**7.3 Specific end use(s)**

Refer to the label and/or leaflet.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1 Control parameters

| Components | CAS-No. | Control parameters | Update | Basis |
|---|---|---|---|---|
| Glufosinate ammonium | 77182-82-2 | 0,9 mg/m3 (TWA) | | OES BCS* |
| 1-Methoxy-2-propanol | 107-98-2 | 375 mg/m3/100 ppm (TWA) | 12 2009 | EU ELV |
| 1-Methoxy-2-propanol | 107-98-2 | 568 mg/m3/150 ppm (STEL) | 12 2009 | EU ELV |
| 1-Methoxy-2-propanol | 107-98-2 | 563 mg/m3/150 ppm (STEL) | 2014 | EU SCOELS |
| 1-Methoxy-2-propanol | 107-98-2 | 375 mg/m3/100 ppm (TWA) | 2014 | EU SCOELS |

*OES BCS: Internal Bayer AG, Crop Science Division "Occupational Exposure Standard"

### 8.2 Exposure controls

**Personal protective equipment**
In normal use and handling conditions please refer to the label and/or leaflet. In all other cases the following recommendations would apply.

**Respiratory protection**

Respiratory protection is not required under anticipated

circumstances of exposure.
Respiratory protection should only be used to control residual risk of short duration activities, when all reasonably practicable steps have been taken to reduce exposure at source e.g. containment and/or local extract ventilation. Always follow respirator manufacturer's instructions regarding wearing and maintenance.
Wear respirator with an organic vapours and gas filter mask (protection factor 10) conforming to EN140 type A or equivalent.

**Hand protection**

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. Also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion, and the contact time.
Wash gloves when contaminated. Dispose of when contaminated inside, when perforated or when contamination outside cannot be removed.

| | |
|---|---|
| Material | Nitrile rubber |
| Rate of permeability | > 480 min |
| Glove thickness | > 4 mm |
| Protective index | Class 6 |
| Directive | Protective gloves complying with EN 374. |

**Eye protection**

Wear goggles (conforming to EN166, Field of Use = 5 or equivalent) and faceshield (conforming to EN166, Field of Use = 3 or equivalent).

**Skin and body protection**

Wear standard coveralls and Category 3 Type 4 suit.
If there is a risk of significant exposure, consider a higher protective type suit.
Wear two layers of clothing wherever possible. Polyester/cotton or cotton overalls should be worn under chemical protection suit and should be professionally laundered frequently.
If chemical protection suit is splashed, sprayed or significantly contaminated, decontaminate as far as possible, then carefully remove and dispose of as advised by manufacturer.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1 Information on basic physical and chemical properties

| | |
|---|---|
| **Form** | Liquid |
| **Colour** | blue to blue green |
| **Odour** | weakly pungent |
| **pH** | 6,8 - 7,8 at 100 % (23 °C) |
| **Flash point** | 57 °C<br>Does not sustain combustion. |
| **Density** | ca. 1,11 g/cm³ at 20 °C |
| **Water solubility** | at 20 °C<br>miscible |

| | |
|---|---|
| **Partition coefficient: n-octanol/water** | Glufosinate-ammonium: log Pow: -4,01 at pH 7<br>Alkylethersulfate, sodium salt: log Pow: 0,3 |
| **9.2 Other information** | Further safety related physical-chemical data are not known. |

## SECTION 10: STABILITY AND REACTIVITY

**10.1 Reactivity**

| | |
|---|---|
| **Thermal decomposition** | > 200 °C, Heating rate: 10 K/min |
| **10.2 Chemical stability** | Stable under normal conditions. |
| **10.3 Possibility of hazardous reactions** | No hazardous reactions when stored and handled according to prescribed instructions. |
| **10.4 Conditions to avoid** | Extremes of temperature and direct sunlight. |
| **10.5 Incompatible materials** | Bases |
| **10.6 Hazardous decomposition products** | Ammonia |

## SECTION 11: TOXICOLOGICAL INFORMATION

**11.1 Information on toxicological effects**

| | |
|---|---|
| **Acute oral toxicity** | LD50 (Rat) 1.730 mg/kg |
| **Acute inhalation toxicity** | LC50 (Rat) 2,97 mg/l<br>Exposure time: 4 h<br>Determined in the form of a respirable aerosol.<br>During intended and foreseen applications, no respirable aerosol is formed. |
| **Acute dermal toxicity** | LD50 (Rat) 593 mg/kg |
| **Skin irritation** | Slight irritant effect - does not require labelling. (Rabbit) |
| **Eye irritation** | Severe eye irritation. (Rabbit) |
| **Sensitisation** | Non-sensitizing. (Guinea pig)<br>OECD Test Guideline 406, Buehler test |

**Assessment STOT Specific target organ toxicity – single exposure**

Glufosinate-ammonium: Based on available data, the classification criteria are not met.

Alkylethersulfate, sodium salt: Based on available data, the classification criteria are not met.

**Assessment STOT Specific target organ toxicity – repeated exposure**

Glufosinate-ammonium caused neurobehavioral effects and/or neuropathological changes in animal studies. Glufosinate-ammonium was well tolerated in rats and mice but less well tolerated in the dog in subchronic studies.

Alkylethersulfate, sodium salt did not cause specific target organ toxicity in experimental animal studies.

**Assessment mutagenicity**

Glufosinate-ammonium was not mutagenic or genotoxic in a battery of in vitro and in vivo tests.
Alkylethersulfate, sodium salt was not mutagenic or genotoxic in a battery of in vitro and in vivo tests.

**Assessment carcinogenicity**

Glufosinate-ammonium was not carcinogenic in lifetime feeding studies in rats and mice.
Alkylethersulfate, sodium salt was not carcinogenic in lifetime feeding studies in rats and mice.

**Assessment toxicity to reproduction**

Implantation loss occurred in a rat multigeneration study with Glufosinate-ammonium. There were no effects on male fertility.
Alkylethersulfate, sodium salt did not cause reproductive toxicity in a two-generation study in rats.

**Assessment developmental toxicity**

Glufosinate-ammonium caused developmental toxicity only at dose levels toxic to the dams.
Glufosinate-ammonium caused an increased incidence of post implantation losses.
Alkylethersulfate, sodium salt did not cause developmental toxicity in rats and rabbits.

**Aspiration hazard**

Based on available data, the classification criteria are not met.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1 Toxicity

| | |
|---|---|
| **Toxicity to fish** | LC50 (Oncorhynchus mykiss (rainbow trout)) 13,4 mg/l<br>Exposure time: 96 h |
| **Toxicity to aquatic invertebrates** | EC50 (Daphnia magna (Water flea)) 17,8 mg/l<br>Exposure time: 48 h |
| **Toxicity to aquatic plants** | ErC50 (Raphidocelis subcapitata (freshwater green alga)) 8,72 mg/l<br>Growth rate; Exposure time: 72 h |
| | NOEC (Raphidocelis subcapitata (freshwater green alga)) 1,25 mg/l<br>Exposure time: 72 h |

### 12.2 Persistence and degradability

| | |
|---|---|
| **Biodegradability** | Glufosinate-ammonium:<br>Not rapidly biodegradable<br>Alkylethersulfate, sodium salt:<br>rapidly biodegradable |
| **Koc** | Glufosinate-ammonium: Koc: 2,3<br>Alkylethersulfate, sodium salt:No data available |

### 12.3 Bioaccumulative potential

| | |
|---|---|
| **Bioaccumulation** | Glufosinate-ammonium: Bioconcentration factor (BCF) < 1<br>Does not bioaccumulate.<br>Alkylethersulfate, sodium salt:<br>Does not bioaccumulate. |

### 12.4 Mobility in soil

| | |
|---|---|
| **Mobility in soil** | Glufosinate-ammonium: Highly mobile in soils<br>Alkylethersulfate, sodium salt: soluble in water |

**12.5 Results of PBT and vPvB assessment**

| | |
|---|---|
| **PBT and vPvB assessment** | Glufosinate-ammonium: This substance is not considered to be persistent, bioaccumulative and toxic (PBT). This substance is not considered to be very persistent and very bioaccumulative (vPvB).<br>Alkylethersulfate, sodium salt: This substance is not considered to be persistent, bioaccumulative and toxic (PBT). This substance is not considered to be very persistent and very bioaccumulative (vPvB). |

**12.6 Other adverse effects**

| | |
|---|---|
| **Additional ecological information** | No further ecological information is available. |

## SECTION 13: DISPOSAL CONSIDERATIONS

**13.1 Waste treatment methods**

| | |
|---|---|
| **Product** | In accordance with current regulations and, if necessary, after consultation with the site operator and/or with the responsible authority, the product may be taken to a waste disposal site or incineration plant. |
| **Contaminated packaging** | Not completely emptied packagings should be disposed of as hazardous waste. |
| **Waste key for the unused product** | **02 01 08*** agrochemical waste containing hazardous substances |

## SECTION 14: TRANSPORT INFORMATION

**ADR/RID/ADN**

| | |
|---|---|
| 14.1 UN number | **2902** |
| 14.2 Proper shipping name | PESTICIDE, LIQUID, TOXIC, N.O.S.<br>(GLUFOSINATE-AMMONIUM SOLUTION) |
| 14.3 Transport hazard class(es) | 6.1 |
| 14.4 Packing group | III |
| 14.5 Environm. Hazardous Mark | NO |
| Hazard no. | 60 |
| Tunnel Code | E |

This classification is in principle not valid for carriage by tank vessel on inland waterways. Please refer to the manufacturer for further information.

**IMDG**

| | |
|---|---|
| 14.1 UN number | **2902** |
| 14.2 Proper shipping name | PESTICIDE, LIQUID, TOXIC, N.O.S.<br>(GLUFOSINATE-AMMONIUM SOLUTION) |
| 14.3 Transport hazard class(es) | 6.1 |
| 14.4 Packing group | III |
| 14.5 Marine pollutant | NO |

| | |
|---|---|
| Segregation group according to 5.4.1.5.11.1 | IMDG SEGREGATION GROUP 2 - AMMONIUM COMPOUNDS |

**IATA**

| | |
|---|---|
| 14.1 UN number | **2902** |
| 14.2 Proper shipping name | PESTICIDE, LIQUID, TOXIC, N.O.S. (GLUFOSINATE-AMMONIUM SOLUTION ) |
| 14.3 Transport hazard class(es) | 6.1 |
| 14.4 Packing group | III |
| 14.5 Environm. Hazardous Mark | NO |

**14.6 Special precautions for user**
See sections 6 to 8 of this Safety Data Sheet.

**14.7 Transport in bulk according to Annex II of MARPOL and the IBC Code**
No transport in bulk according to the IBC Code.

## SECTION 15: REGULATORY INFORMATION

**15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture**

**Further information**

WHO-classification: II (Moderately hazardous)

**15.2 Chemical safety assessment**

A chemical safety assessment is not required.

## SECTION 16: OTHER INFORMATION

**Text of the hazard statements mentioned in Section 3**

| | |
|---|---|
| H226 | Flammable liquid and vapour. |
| H302 | Harmful if swallowed. |
| H312 | Harmful in contact with skin. |
| H315 | Causes skin irritation. |
| H318 | Causes serious eye damage. |
| H332 | Harmful if inhaled. |
| H336 | May cause drowsiness or dizziness. |
| H360Fd | May damage fertility. Suspected of damaging the unborn child. |
| H373 | May cause damage to organs through prolonged or repeated exposure. |
| H412 | Harmful to aquatic life with long lasting effects. |

**Abbreviations and acronyms**

| | |
|---|---|
| ADN | European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways |
| ADR | European Agreement concerning the International Carriage of Dangerous Goods by Road |
| ATE | Acute toxicity estimate |
| CAS-Nr. | Chemical Abstracts Service number |

| | |
|---|---|
| Conc. | Concentration |
| EC-No. | European community number |
| ECx | Effective concentration to x % |
| EINECS | European inventory of existing commercial substances |
| ELINCS | European list of notified chemical substances |
| EN | European Standard |
| EU | European Union |
| IATA | International Air Transport Association |
| IBC | International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk (IBC Code) |
| ICx | Inhibition concentration to x % |
| IMDG | International Maritime Dangerous Goods |
| LCx | Lethal concentration to x % |
| LDx | Lethal dose to x % |
| LOEC/LOEL | Lowest observed effect concentration/level |
| MARPOL | MARPOL: International Convention for the prevention of marine pollution from ships |
| N.O.S. | Not otherwise specified |
| NOEC/NOEL | No observed effect concentration/level |
| OECD | Organization for Economic Co-operation and Development |
| RID | Regulations concerning the International Carriage of Dangerous Goods by Rail |
| TWA | Time weighted average |
| UN | United Nations |
| WHO | World health organisation |

The information contained within this Safety Data Sheet is in accordance with the guidelines established by Regulation (EU) 1907/2006 and Regulation (EU) 2015/830 amending Regulation (EU) No 1907/2006 and any subsequent amendments. This data sheet complements the user's instructions, but does not replace them. The information it contains is based on the knowledge available about the product concerned at the time it was compiled. Users are further reminded of the possible risks of using a product for purposes other than those for which it was intended. The required information complies with current EEC legislation. Addressees are requested to observe any additional national requirements.

Changes since the last version are highlighted in the margin. This version replaces all previous versions.